MILITARY SPECIFICATION

DETERGENT, LAUNDRY, NONIONIC, LIQUID

This specification is approved for use by Natick Research and Development Command, Department of the Army, and is available for use by all Deparments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a liquid nonionic synthetic detergent suitable for laundering cotton, synthetic, and woolen fabrics in soft, hard or sea water; using a liquid supply dispensing system.

1.2 Classification. The detergent covered by this specification shall be of one type and shall be furnished in the following capacity containers, as specified (see 6.2).

5 gallons (18.9 liters)
15 gallons (56.8 liters)
55 gallons (208 liters)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680 - Dry Cleaning Solvent

MILITARY

MIL-D-43703 - Drums, Shipping and Storage, Molded Polyethylene

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7930

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STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing
FED-STD-536 - Soap and Soap-Products (Including Synthetic Detergents) Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners or Pallet-Type Base

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

16 CFR 1500 - Federal Hazardous Substance Act

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or requests for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Classification Committee

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The liquid laundry detergent shall be a clear and homogeneous blend of synthetic detergents, emulsifiers and soil suspending agents. The product shall not contain phosphates, carbonates, silicates, or organic solvents. It shall conform to the requirements in table I, when tested as specified in 4.3.2.

TABLE I. Chemical and physical requirement

Property	Requirement
pH (1% solution)	7.5 - 9.5
Cloud point at 1% in distilled water	74°C minimum
Sudsing, ml of foam at 38°C	30 maximum
Specific gravity at 25/25°C	1.000 to 1.020
Nonionic surfactant	30% by weight minimum
Fabric brightener content	0.1 %
Organic chelating agent	Trace
Organic soil suspending agent	2.5 % by weight, minimum
Odor	
Initial	No kerosene - like, rancid, fishy or acrid odor
Residual	
Appearance	Clear viscous liquid at 25°C

3.1.1 Certificate of compliance. The contractor shall submit to the contracting officer a certificate of compliance stating that no enzyme additives, derived from spore-forming bacteria, in excess of residual amounts found in raw materials have been made a part of formulations provided under this specification.

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3.2 Nonionic surfactant. The nonionic surfactant in the liquid laundry detergent shall consist of either linear alcohol ethoxylates or alkyl aryl polyether alcohol types with optional fatty amides present or mixtures thereof. The surfactant shall meet the following requirements (see 4.2.1).

Properties of Nonionic surfactant	Min.	Max.
Molecular weight (average)	525	650
Specific gravity (25°C)	0.950	1.07
Cloud point (1% aqueous solution)	125°F (51°C)	170°F (77°C)
pH, 1% solution (at 25°C)	6	8
Water, percent	--	0.3
Flash point °C	204	--
Biodegradable, percent	90	--
Alkyl carbon range	C8	C15
Average moles of ethylene oxide	6	10

3.3 Labeling. Each container of detergent shall be durably and legibly marked with the following information, precautions, and directions for use and shall conform to the Federal Hazardous Substances Act:

Detergent Laundry Nonionic Liquid.

A mild, low sudsing liquid laundry detergent containing a special blend of synthetic detergents, emulsifiers, and soil suspenders for use in laundering synthetic fabrics, woolens, and cottons in soft, hard or sea water, in field and fixed laundry wash wheels. It may be used in conjunction with an alkaline builder in liquid supply dispensing systems. When preparing stock solutions for use in liquid systems, "warm, soft water" should be used.

CAUTION: DO NOT USE IN DISHWASHING MACHINES OR ON AIRCRAFT SURFACES. DO NOT TAKE INTERNALLY. KEEP OUT OF EYES. IF SWALLOWED, INDUCE VOMITING AND CALL A PHYSICAN. FLUSH EYES WITH WATER.

Soil Classification	Amount Used Per 100 pound Load
Light	2-3 ounces (57-85 g)
Medium	4 ounces (113 g)
Heavy	6-8 ounces (170-227 g)

3.4 Workmanship. The detergent shall be free of foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examinations and tests shall consist of all the end items prepared in a single batch and submitted for inspection at one time.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements referenced specifications and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents. In addition, the contractor shall furnish a certificate of compliance stating that the nonionic surfactant conforms to the requirements of 3.2.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs below at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.3. The lot size for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.3.1.1 and in units of palletized unit loads in 4.3.1.3. The end item lot shall consist of all of the detergent manufactured in one batch.

4.3.1.1 Packaging inspection. An examination shall be made to determine that packing and marking requirements of Section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it shall not be palletized. Containers fully prepared for delivery that have not been palletized shall be examined for closure defects.

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Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Marking	Omitted; incorrect; illegible; of improper size, location or sequence. Not as specified.
Material	Component missing, damaged or otherwise defective.

4.3.1.2 Examination for palletization. An examination shall be made to determine that the palletization complies with the Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot submitted for inspection at one time.

TABLE II. Examination of palletization

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.3.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	S-2	2.5
4.3.1.2	S-1	6.5

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4.3.2 Testing of the end item. The end item shall be tested for the characteristics specified in table I. The number of sample units for composite testing shall be as specified in 4.3.2.1. The lot size for test purposes shall be the number of gallons of detergent submitted for inspection at one time. The sample size shall be shown below. Tests shall be performed on a composite sample. The lot shall be unacceptable if the composite sample fails to meet any test requirement specified.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.3.2.1 Composite sample. The sample unit for composite testing shall be 1 pint of detergent selected in equal portions from each sample unit. The portions from the individual samples shall be thoroughly mixed, yielding the composite. The composite sample unit shall be placed in an airtight and watertight glass container with suitable caps, covered and sealed to prevent any moisture change.

TABLE III. Instructions for testing of the end item.

Characteristic	Specification reference		Number of determinations per sample unit	Results reported as	
	Requirement	Test method		Pass or fail	Numerically to nearest
pH (1% solution)	Table I	4.4.1	Avg of 2	-	0.1 unit
Cloud point	Table I	4.4.2	1	-	1°C
Sudsing	Table I	4.4.3	1	-	1 percent
Specific gravity	Table I	4.4.4	1	-	0.001
Nonionic surfactant	Table I	1/	-	-	-
Organic soil suspending agent	Table I	1/	-	-	-
Organic chelating agent	Table I	1/	-	-	-

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TABLE III. Instructions for testing of the end item (cont'd)

Characteristic	Specification reference: Requirement	Specification reference: Test method	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Fabric brightener	Table I	1/	-	-	-
Nonionic surfactant	Table I	1/	-	-	-
Odor					
Initial	Table I	4.5.1	1	X	-
Residual	Table I	4.5.2	1	X	-
Appearance	Table I	Visual	1	X	-

1/ The contractor shall furnish a certificate of compliance stating that his product meets this requirement.

4.4 Test procedures.

4.4.1 pH determination. Dissolve 1.0 g of the detergent in freshly boiled and cooled distilled water, dilute to 100 ml with freshly boiled and cooled distilled water and mix. Determine the pH value (to the nearest 0.1) of this solution at 25°C by means of a suitable pH meter employing a glass electrode. Do not correct the results for sodium ion concentration.

4.4.2 Cloud point. Dissolve 1.0 g of detergent in 100 ml of distilled water. Transfer 10 ml of this solution to a clean, dry 25 x 200 mm test tube, and insert a thermometer. The test tube shall then be inserted about halfway in an 800-ml beaker nearly full of water. Mechanical stirring of the water shall be provided and low heat shall be applied to raise the temperature slowly. The sample shall not be stirred until a preliminary indication of the cloud point is obtained by turbidity developing in the layers next to the wall of the test tube. From the time this is observed until the solution suddenly becomes cloudy, the sample shall be stirred continuously by the thermometer. The cloud point shall be reported in °F (°C) as the temperature at which cloudiness appears throughout.

4.4.3 Sudsing. Determine the sudsing property of the detergents in accordance with Method 2001 of FED-STD-536, with the following modification:

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ADD 2.0 ± 0.01 grams of detergent in 1 liter of distilled or deionized water at 38 ± 1°C taking care not to generate a foam. Immerse (for 10 minutes) a 250 ml unstoppered graduated cylinder up to the 150 ml graduation mark in a water bath maintained at 38 ± 1°C. Carefully pour 50 ml of the detergent solution into the preheated graduated cylinder and allow to remain in the bath for 2 minutes prior to shaking. Stopper the graduated cylinder and hold the stopper to prevent the stopper from loosening due to internal pressure build up, and proceed as directed in FED-STD-536.

4.4.4 Specific gravity. The specific gravy shall be taken in accordance with method 4183 of FED-STD-141. Test 5, Specific Gravity by Means of a Pycnometer, shall be used. The specific gravity shall be determined at 25°C.

4.4.5 Odor.

4.4.5.1 Initial. The odor of the sample shall not be kerosene-like, rancid, fishy or acrid.

4.4.5.2 Residual. Allow a 4-inch by 4-inch (10 cm by 10 cm) swatch of white wool cloth (see 6.3) to stand in a 0.75 percent by volume solution of the detergent in type I solvent specified in P-D-680 for 30 minutes. Remove the swatch without wringing, allow to drain on a wire screen for two minutes, place in a wide mouth bottle and shake with 200 ml. of the solvent for 5 minutes. Repeat the rinse for another 5 minutes using fresh solvent and then hang the swatch to dry at 24 ± 2°C overnight. Place in an oven at 38 ± 1°C for 2 hours. The sample shall be examined for odors specified in table I before and after placing in the oven. Kerosene-like, rancid, fishy or acrid odors present after oven drying shall constitute failure.

5. PACKAGING

5.1 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Detergent shall be furnished in five, fifteen and fifty-five gallon, self-supporting, molded polyethylene drums, conforming to MIL-D-43703. Ultraviolet light protection is not required for product protection. The five gallon and fifteen gallon containers shall be provided with a reclosable dispenser feature which is of such size and shape to allow the detergent to be dispensed from the container.

5.1.2 Commercial. Detergent shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The contents per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.2 Palletization. When specified (see 6.2), detergent packed as specified in 5.1 shall be palletized in accordance with load type XII or XIII (as applicable) of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the Contracting Officer.

5.3 Marking. In addition to any special marking required by the contract or order, containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The detergent is suitable for laundering cotton, wool and synthetic fabrics in soft, hard or sea water.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size container required (see 1.2).
(c) Selection of applicable level of packing (see 5.1).

6.3 Wool cloth. Cloth, wool, 100 percent virgin, unbleached. Approximately 47 ends and 41 picks per inch and approximately 6 ounces per square yard (203 g/m^2).

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-43966(GL) Detergent, Laundry, Nonionic, Liquid

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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